# Zhongji Innolight (300308.SZ)

*Current edition | Company valuation methods and portfolio Base scores preserved.*

**CURRENT EDITION - 1 October 2026 | Framework 8.0**

Full investment research, current probability-weighted operating model and analytical rescore. Financial assumptions, source evidence and forward estimates are distinguished throughout.

## Q4 2027 ceiling

| Current model item | Value | |
|---|---|---|
| Q4 2027 ceiling | | 1,027.2202 CNY/share |
| Valuation window | Q4 2027 to Q3 2028 | |
| Artifact analytical Base | | 72/100 |
| Protected portfolio Excel Base | | 72/100 |

Qualified components become more expensive inside part of the advanced-product pool. Customers absorb half the increase. There is no blanket-ban revenue recovery to add.

The sanctioned valuation remains **20x forward-four-quarter operating profit**, with the existing accounting basis, ownership, shares and currency conversion. The active ceiling next rolls on January 1, 2027.

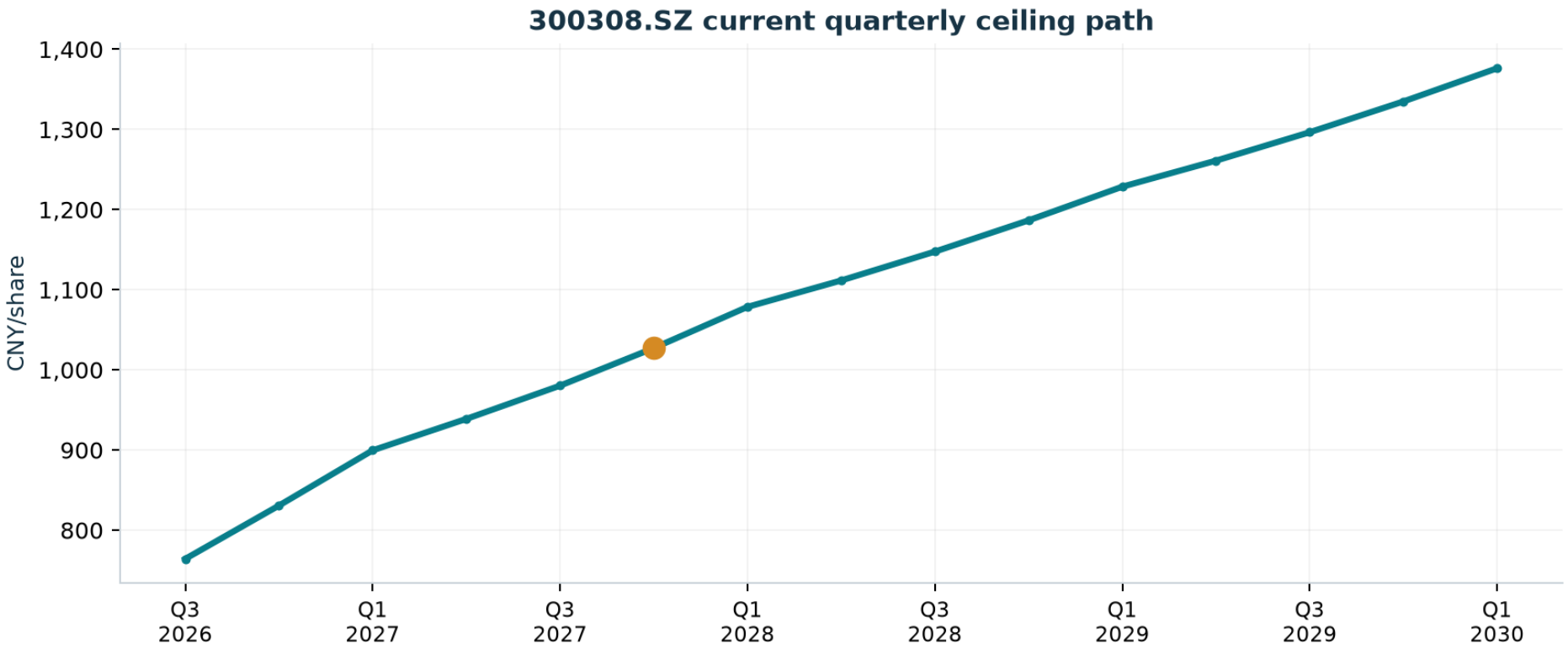



*Figure 1. Current quarterly ceilings, CNY/share. Gold marks the active Q4 2027 horizon. The 40% policy contribution is included once in operating earnings.*

| Ceiling quarter | Current ceiling CNY/share |
|---|---|
| Q3 2026 | 764.2155 |
| Q4 2026 | 830.7502 |
| Q1 2027 | 899.6356 |
| Q2 2027 | 938.8624 |
| Q3 2027 | 980.3518 |
| Q4 2027 | 1,027.2202 |
| Q1 2028 | 1,078.5769 |
| Q2 2028 | 1,111.5605 |

| Ceiling quarter | Current ceiling CNY/share |
|---|---|
| Q3 2028 | 1,147.5533 |
| Q4 2028 | 1,186.5443 |
| Q1 2029 | 1,228.5332 |
| Q2 2029 | 1,261.0329 |
| Q3 2029 | 1,296.4860 |
| Q4 2029 | 1,334.8923 |
| Q1 2030 | 1,376.2520 |

## Analytical Base and scored conviction

| Category | Score | Evidence and limit |
|---|---|---|
| Core Thesis | 16/20 | Qualified scale and yields; alternatives prevent monopoly credit. |
| Cycle Stage | 9/10 | 1.6T shipment mix and funded customer expansion; inputs constrain delivery. |
| TAM Expansion | 7/10 | Large DB1 expansion; adjacent markets not separately contracted. |
| Catalysts | 4/5 | H2 delivery, new factories in 2027, advanced products H2 2027. |
| Macro | 9/10 | AI networking intensity and customer capex; policy exposure persists. |
| Alpha | 2/5 | Chinese Q&A and cash-flow reconciliation add insight; widely followed leader. |
| Risks | -4 | Competition; execution/working capital; customer concentration; policy/FX. |
| Total | 43/60 = 72/100 | Watchlist; price-independent analytical thesis score |

**Artifact analytical Base: 72/100. Protected portfolio Excel Base: 72/100.** The portfolio value is a separate retained decision; it is not a scoring-table input.

The narrower policy scenario does not restore revenue that was never removed for a blanket ban. Qualified-component inflation and partial customer pass-through affect profit directly. Existing policy, execution and competition risks already cover the uncertain sourcing environment; thesis-quality categories remain unchanged.

Base measures thesis quality. The 40% policy probability weights only the incremental operating economics, once; neither Base nor the policy weight is applied again to the final ceiling. Scores use integer category points on the /60 framework lattice.

# Policy breadcrumb and economic hypothesis

Serenity's October 1 post attributes Washington meetings to Morgan Stanley and reports a possible 3.2T-focused restriction, with a potential 65% U.S.-value route for Chinese modules. The source expects laser pricing to matter more than broad module-share redistribution. Treat this as an attributed policy scenario; the original MS note and final FCC language establishing these exact terms were not recovered. The separate September 25 Senate proposal concerns sensitive federal systems and a five-year transition. It is not confirmation of these FCC terms.

Serenity policy post | Foundry follow-up | Separate Senate proposal

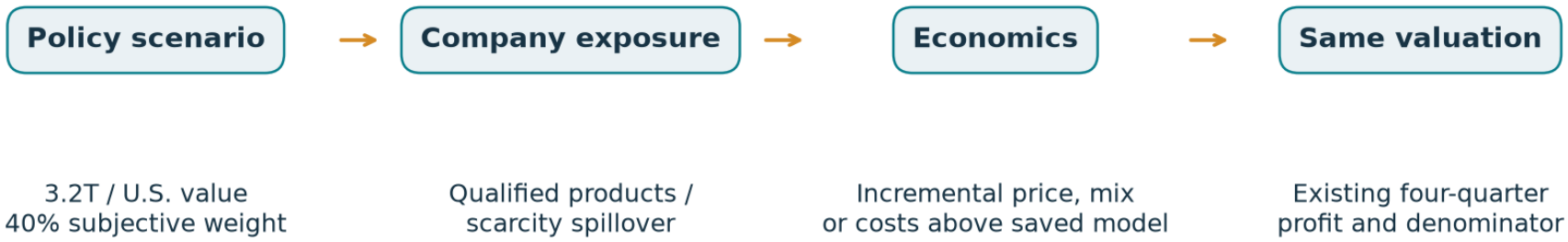



*Figure 2. Analytical transmission of the attributed policy scenario. This is a model diagram, not an official regulatory diagram or customer award.*

**Assigned common policy-scenario probability: 40%.** Subjective initial weight above the single-source 25-35% band because Serenity has the owner's highest weighting and attributes Washington meetings to Morgan Stanley. The exact future rule, timing and economic effect remain uncertain. Serenity and Morgan Stanley are one source chain, not two independent confirmations.

Half the existing probability-weighted advanced pool is a bounded 3.2T-content proxy. Repricing 40% of that product COGS by 5%, with half passed through, isolates incremental component costs. No blanket-ban recovery is booked because no such loss exists in the reference model.

Eligible fractions are bounded scenario assumptions, not origin certificates. U.S. corporate domicile and a U.S. listing do not prove manufacturing origin. If the final content definition excludes the modeled products, their eligible fraction goes to zero.

For component and integrated suppliers, the filtered revenue pool means 3.2T-linked sales plus incremental cross-generation pricing spillover from competition for common qualified capacity. It is not a claim that 800G/1.6T is directly regulated. Fractions are LOW-confidence exposure proxies, not verified geographic market shares.

| Signal ledger field | Current treatment |
|---|---|
| Signal ID / date | FCC-OPTICS-20261001; Serenity/MS source chain |
| What is already included | All saved product growth, shortages, capacity, customer probabilities and contracts. The bridge adds only the explicitly calculated incremental economics. |
| Probability and timing | 40% common policy scenario; price/cost intensity 0 before Q3 2027, 25% Q3 2027, 50% Q4 2027, 75% Q1 2028, 100% from Q2 2028. AAOI mix uses its own slot ramp. |
| Numerical contribution | Q4 2027 operating quarter: delta revenue 0.633213, delta consolidated OP -0.666457 (CNY millions). Forward-four-quarter valuation OP changes -8.737325. |
| Dependence | The policy probability is shared across companies. These are correlated portfolio exposures, not independent confirmations. Existing customer-conversion probabilities are already inside source revenue. |
| Raise weight | Rule/proposal scope supports qualifying content and timing; company product/foundry evidence improves; customer sourcing or like-for-like price negotiations support the incremental premium. |
| Reduce weight | Policy is delayed beyond the horizon, the content definition excludes products, supply eases, fixed contracts prevent repricing, unit costs absorb the premium, or 3.2T conversion slips. |
| Zero-contribution boundary | No broad TAM uplift, no new factory capacity, no OCS benefit and no blanket-ban recovery. AAOI receives no modeled units before 2028. MXL is assessed separately as a candidate. |

## Conditional inputs selected before calculating targets

| Input | Value | Evidence status |
|---|---|---|
| 3.2T share of advanced pool | 50.00% | LOW-confidence analyst assumption; inherited cap where labeled |
| Affected qualifying-component COGS fraction | 40.00% | LOW-confidence analyst assumption; inherited cap where labeled |
| Component unit-cost increase | 5.00% | LOW-confidence analyst assumption; inherited cap where labeled |
| Component-cost increase passed to customers | 50.00% | LOW-confidence analyst assumption; inherited cap where labeled |

Incremental COGS and opex rates apply to incremental price revenue, not original product revenue or COGS. AAOI mix cost uses unit COGS separately. AXTI stranded COGS is a fraction of COGS attached to lost units. Chinese component-cost rates apply to affected reference COGS.

**Timing is an analyst sensitivity.** A deliberately labeled early-sourcing proxy ahead of a 2028 3.2T ramp. Earlier quarters receive zero incremental policy revenue. Full intensity begins Q2 2028 and remains an uncertain continuation through the saved forecast only. These dates are not announced orders or a published regulatory timetable. The probability sensitivity below includes a no-policy control; a later start would reduce the near-term contribution.

# Incremental revenue, cost and profit

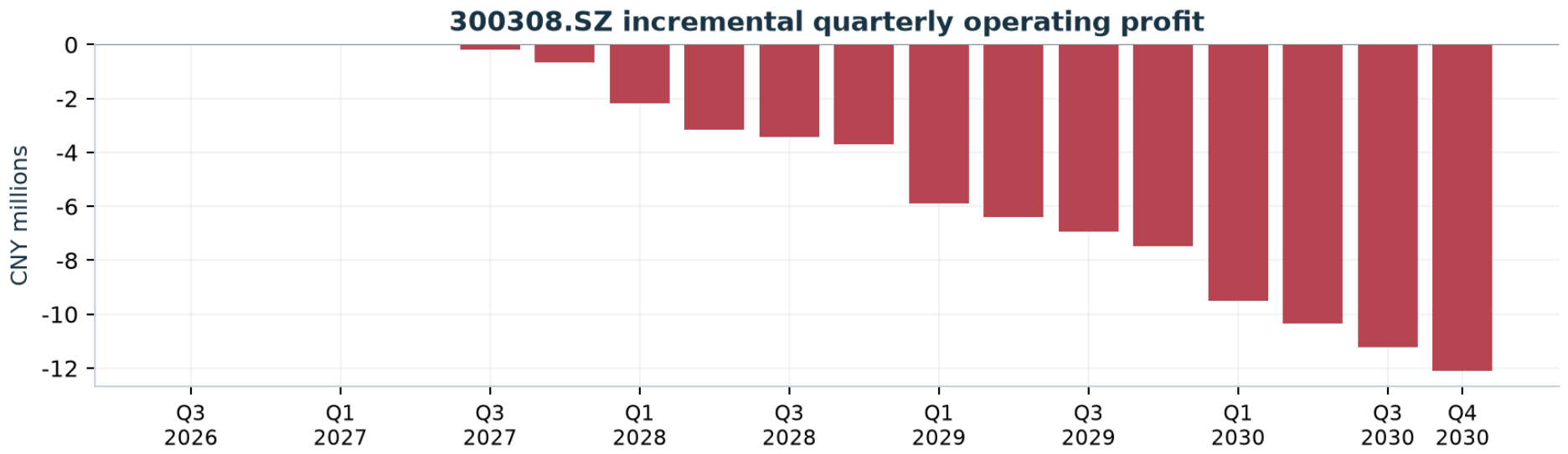



*Figure 3. Incremental consolidated operating profit, CNY millions. Negative bars represent the net weighted adverse policy effect; baseline operations are unchanged.*

| Quarter | Delta revenue | Delta COGS | Delta gross profit | Delta opex | Delta OP |
|---|---|---|---|---|---|
| Q3 2026 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| Q4 2026 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| Q1 2027 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| Q2 2027 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| Q3 2027 | 0.1764 | 0.3528 | -0.1764 | 0.0093 | -0.1857 |
| Q4 2027 | 0.6332 | 1.2664 | -0.6332 | 0.0332 | -0.6665 |
| Q1 2028 | 2.0699 | 4.1398 | -2.0699 | 0.1045 | -2.1744 |
| Q2 2028 | 3.0107 | 6.0215 | -3.0107 | 0.1520 | -3.1628 |
| Q3 2028 | 3.2616 | 6.5233 | -3.2616 | 0.1647 | -3.4264 |
| Q4 2028 | 3.5125 | 7.0251 | -3.5125 | 0.1774 | -3.6899 |
| Q1 2029 | 5.5963 | 11.1926 | -5.5963 | 0.2826 | -5.8789 |

| Quarter | Delta revenue | Delta COGS | Delta gross profit | Delta opex | Delta OP |
|---|---|---|---|---|---|
| Q2 2029 | 6.1051 | 12.2101 | -6.1051 | 0.3083 | -6.4134 |
| Q3 2029 | 6.6138 | 13.2276 | -6.6138 | 0.3340 | -6.9478 |
| Q4 2029 | 7.1226 | 14.2451 | -7.1226 | 0.3597 | -7.4823 |
| Q1 2030 | 9.0314 | 18.0628 | -9.0314 | 0.4651 | -9.4965 |
| Q2 2030 | 9.8524 | 19.7049 | -9.8524 | 0.5074 | -10.3598 |
| Q3 2030 | 10.6735 | 21.3469 | -10.6735 | 0.5497 | -11.2231 |
| Q4 2030 | 11.4945 | 22.9890 | -11.4945 | 0.5920 | -12.0865 |

### Active-window product bridge

| Product / event | Pre-policy operating pool | Filtered exposure | Weighted delta revenue | Weighted delta OP |
|---|---|---|---|---|
| Advanced-component costs and partial pass-through | 1,194.7421 | 126.6427 | 0.6332 | -0.6665 |

Filtered exposure is a scenario allocation, not verified U.S.-origin revenue. For cost cases the exposure is an affected cost pool; AAOI substitution units are explained in its separate bridge. All financial amounts use the company table units.

## Business research and source evidence

The following sets out the company thesis, historical results, guidance, capacity, program probabilities, risks and original source images. Source dates remain visible. Operating assumptions below feed the complete current financial tables, including the explicit probability-weighted policy contribution.

### 1. Thesis, classification and score

Innolight converts electrical signals into optical signals at scale for AI and cloud networks. Its advantage combines customer qualification, yield, manufacturing scale, product engineering and access to scarce DSPs and lasers. It is a leading beneficiary with temporary delivery advantages; it is not an irreplaceable laser or DSP monopoly. Optical links are necessary, but customers can qualify competing module suppliers.

The investable operating inflection is the 1.6T mix shift already visible in H1, followed by customer-specific NPO and higher-content products in 2027-28. The critical test is whether demand becomes shipped modules with cash collection while retaining gross margin. Adding capacity alone does not establish sales. The July and August management records describe constrained inputs, substantial supply commitments and later production ramps; those disclosures outweigh a generic industry shipment forecast [S1,S4,S5].

The S-curve is high-volume 800G, scaling 1.6T, and early qualification for advanced optics. Phase 1 AI/datacenter connectivity is established. Phase 2 edge inference and industrial networks can extend optical traffic, but no distinct incremental customer wallet is verified. Phase 3 ubiquitous optical compute is engineering optionality,

not a second revenue pool. TAM scores 7 rather than a universal-primitive 9 or 10: most monetization remains in one infrastructure ecosystem.

The current summing score is shown in the analytical Base chapter above.

## 2. Reported anchor and accounting controls

H1 and Q1 are consolidated Chinese Accounting Standards reports. Q2 is H1 less Q1, not a standalone company forecast. The A-share CAS statements are the model anchor; the H-share IFRS presentation has different cost classifications and must not be spliced into CAS margins.

| CNY M | Q1 2026 actual | Q2 2026 derived | H1 2026 actual |
|---|---|---|---|
| Revenue | 19,496.40 | 22,281.46 | 41,777.86 |
| Gross profit | 8,979.68 | 10,343.50 | 19,323.17 |
| R&D expense | 644.69 | 508.34 | 1,153.03 |
| Selling + administration | 487.98 | 436.52 | 924.50 |
| Taxes/surcharges + impairments | 107.22 | 83.96 | 191.18 |
| Core OP, derived | 7,739.79 | 9,314.67 | 17,054.46 |
| CAS reported operating profit | 7,535.02 | 9,696.30 | 17,231.32 |
| Parent net income | 5,734.50 | 7,916.65 | 13,651.15 |
| Operating cash flow | 3,367.57 | -1,567.90 | 1,799.67 |

Core OP = sales - COGS - selling - administration - R&D - business taxes/surcharges - credit/asset impairment charges. Depreciation and stock compensation remain expenses. H1 reconciliation: core OP 17,054.46 - finance expense 562.25 + other income 13.89 + investment income 189.52 + fair-value gains 537.12 - disposal loss 1.42 = reported operating profit 17,231.32. Finance expense contains 708.79 of FX losses, partly offset by interest income. The core model excludes both investment gains and net finance/FX effects; zero future FX is an assumption, not reported guidance.

H1 total net income was 14,733.43, of which 1,082.28 belonged to minority interests. The model attributes 92.6543% of core OP to parent shareholders, using parent/total net income as a disclosed proxy because segment-level minority operating profit is unavailable. This is not a legal ownership percentage and can shift with the subsidiary profit mix. Multiplying consolidated OP by 20 without this adjustment would overstate parent economics.

Optical-module revenue was 41,355.21 in H1, versus 37,456.52 for all of 2025. Reported H1 module gross margin was 46.59%, up from 42.61% in FY25. H1 sales of 18.99M modules imply CNY 2,177.74 average module revenue; FY25 21.09M imply CNY 1,776.03. These averages include all speeds and cannot reveal the undisclosed 800G/1.6T mix. H1 stated capacity was 22.15M modules and production 18.86M [S1,S3].

Cash conversion is the main counterweight to the profit ramp: H1 CFO 1,799.67 less capex cash 4,802.16 gives a simple CFO-minus-capex proxy of -3,002.49. Inventory reached 19,825.91; other noncurrent assets rose to 1,784.11 and include supply/equipment prepayments. The H1 cash balance 6,397.13 predates the H-share fundraising and September buyback. It is not a current net-cash figure. Supplier prepayments are cash commitments by Innolight, not prepaid orders from customers [S1,S4].

The July H listing and August greenshoe produce 1,177.909641M total shares. September repurchases of approximately 5.6531M are intended for employee incentive/ownership plans, not cancellation [S6,S7]. They are retained in the economic share pool. A fixed 1,190M forward diluted denominator allows roughly 1.03% additional dilution, close to the H1 basic/diluted EPS gap. It is a model reserve, not an issued count or a new fundraising forecast. No automatic future buyback benefit is assumed.

## 3. China Data Packet and demand conversion

| Variable | Best evidence and use | Confidence |
|---|---|---|
| TAM | DB1 GS global module market: USD67.7B / 131.4B / 148.5B in 2026/27/28. No separate addition of CPO or PhotonLink SAM. [S9,S10,S20] | MEDIUM: dated broker forecast |
| Demand | H1 module sales and capacity plus company July/August customer order commentary; customer identities and spend allocations withheld. [S1,S4,S5] | Actual HIGH; projections MEDIUM/LOW |
| Margin | CAS optical GM 46.59%; management expects stability/modest improvement rather than a large jump. [S1,S4] | HIGH actual; MEDIUM near-term model |
| Timing | Current 1.6T; new capacity mainly 2027; NPO/higher-content ramps H2 2027, larger in 2028. [S4,S5] | MEDIUM |
| Risk | DSP/CW/PCB constraints, prepayments, concentrated customers, multiple architectures and overseas execution. [S1,S3,S4] | Material, included in score |

Customer names and individual wallet shares are not disclosed: assigning the anonymized top customers to Microsoft, Amazon, Alphabet or Meta would fabricate evidence. The FY25 top five represented 75.98% of sales. AH-12 therefore uses the AH-13 Plan-B aggregate spending proxy below. The four hyperscalers are a demand reference basket, not a claimed complete customer list. Spending includes buildings and non-optical assets, with different lease conventions; the effective capture coefficient absorbs those differences and is not a literal share of any one customer's bill of materials.

| USD B capex reference | CY2025 actual/rounded | CY2026 guidance midpoint |
|---|---|---|
| Microsoft incl. finance leases | 118.00 | 175.00 |
| Amazon cash capex | 128.30 | 220.00 |
| Alphabet capex | 91.40 | 200.00 |
| Meta incl. lease principal | 72.22 | 137.50 |
| Total | 409.92 | 732.50 |

**Inherited reference assumptions.** Sources S12-S19; Microsoft's 2025 total is 21.4 + 24.2 + 34.9 + 37.5 across its four calendar quarters. Its new 175B outlook partly reflects moving finance leases to operating leases, not reduced physical investment. Amazon and Alphabet current guidance is sourced to AP/Reuters accounts of the calls, rather than falsely described as independently reviewed full transcripts. 2027 onward is our assumption: each customer's budget +25% in 2027, +20% in 2028, +15% in 2029 and +12% in 2030. No company has guided these exact later figures.

**Inherited reference assumptions.** Historical calibration: 409.92B USD x 7.00 CNY/USD x **1.305360% effective capture** = 37,456.52M module sales, exactly FY25 actual. That is a fitted calibration, not independent proof of future capture. Model demand = annual basket x 7.00 x calibrated capture x optical-content multiplier. Content multipliers versus FY25 are 1.45 / 1.70 / 1.75 / 1.80 / 1.85 for 2026-30. They reflect faster links and more network content per compute dollar described by management, but the magnitudes are analyst assumptions. FX 7.00 is a constant modeling convention, not a current spot claim. Both FX and capture must be refreshed at actual results.

**Inherited reference assumptions.** 2026 H1 actual optical revenue is locked; the remaining modeled annual optical demand is allocated 48% to Q3 and 52% to Q4. Later years allocate 22/24/26/28% to Q1/Q2/Q3/Q4. This expresses sequential capacity ramping, not quarterly guidance. Advanced optics is deducted from the same demand budget before probability weighting; no failed qualification revenue is reassigned to pluggables. Capture/content and timing are the two largest uncertainties. A 10% basket or content reduction cuts the forecast optical revenue by 10%, before any cost response.

## 4. Quarterly revenue vectors and margin assumptions

**Inherited reference assumptions.** V1 legacy/<=400G, V2 800G and V3 1.6T are MEDIUM sourced extrapolations through 2027 and LOW thereafter. V4 advanced optics combines higher-speed/new form-factor and NPO programs; its scale is LOW confidence even though direction/timing has management support. V5 other businesses extrapolates H1 at 5% annual growth, LOW. Speed revenue shares and product margins below are synthetic allocations calibrated to disclosed total/optical margins, not reported segment data.

**Inherited reference assumptions.** Other operating charges are 0.45% of sales, covering business taxes and ordinary impairments. Product margins include ordinary production depreciation; R&D and SG&A include stock compensation. There is no EBITDA add-back. Gross margins ease on established speeds over time; advanced mix offsets part of erosion. Absolute R&D continues to rise despite near-term ratio leverage. These assumptions remain sensitive to component prices, qualification yields and new-factory depreciation.

**Inherited reference assumptions.** Revenue allocation audit: Q3/Q4 2026 optical shares are 10/50/40/0 and 10/44/46/0% across legacy/800G/1.6T/advanced. Q1-Q4 2027 legacy shares are 8/7.5/7/6.5%; 800G 43/40/37/34%; advanced 0/0/3/5%; 1.6T is the residual. 2028 shares are 5/25/60/10%; 2029 4/18/62/16%; 2030 3/12/63/22%. Advanced revenue is then multiplied once by 60% in 2027, 70% in 2028 and 75% thereafter. V5 is outside this optical allocation.

ASP/unit cross-check uses the supplied GS chart approximately: 800G USD420/400/320 and 1.6T USD800/750/600 in 2026/27/28. These are market averages, not Innolight prices. At FX7, inferred shipments are feasibility tests rather than orders. Pricing changes by reach, EML versus silicon photonics and DR versus FR; a single ASP cannot identify true product volume.

**Inherited feasibility comparison.** The 2027 figures are well below the user-supplied 90-100M demand per speed; that demand estimate is not a shipment ceiling guaranteed to every vendor. The 60-70M DSP estimate in the same screenshot has ambiguous product/vendor scope and is not assigned wholly to Marvell or wholly to 1.6T. H1 manufacturing capacity cannot be treated as an annual fixed limit because management is expanding, but no verified quarterly sellable-capacity schedule exists. A numerical capacity cap has therefore not been invented. Future revenue is conditional on expansion and component allocation; shipment/capacity reconciliation is a mandatory refresh trigger.

## 5. Valuation method and accounting convention

Model values, CNY M. The current operating chapter distinguishes consolidated OP and its operating margin from the separate Valuation OP column, which applies the inherited owner-attribution proxy. Modeled margins are not reported CAS margins. The last three quarters support the Q1 2030 four-quarter window and do not receive separate live ceilings.

Every ceiling = the sum of the four named quarterly parent core OP figures x 20 / 1,190M shares. The opening quarter is included. These are framework ceilings, not discounted present values or broker targets. The further out the table extends, the less reliable it is; the next 12-16 months receive priority.

No separate terminal TAM is manufactured for 2029-30. The extrapolated capture must be retested against a future market update; it is LOW confidence. Mature 800G/1.6T ASP erosion and new-form-factor substitution can lower revenue even if bits shipped rise.

No excess cash is separately added under the framework ceiling convention; a transaction EV bridge would need current cash/debt after the H listing and buyback.

## 6. Cross-book and risks

Innolight and Eoptolink are competing qualified assemblers exposed to the same hyperscaler spend, so owning both does not create independent end-demand. AAOI and Coherent overlap in transceivers; Coherent PhotonLink adds an integration competitor. MRVL/AVGO DSP supply and LITE/COHR/SIVE CW laser qualification are paired cross-watches. A laser shortage benefits qualified merchant suppliers but can delay module shipments; it is not uniformly bullish across every layer. SOI, InP substrates and packaging suppliers may gain from silicon-photonics adoption without winning a specific module vendor's design.

The four scored risk classes are: (1) competition/architecture, including price erosion and bundled platforms; (2) execution and working-capital funding, including component allocation, overseas ramp yields and customer collections; (3) customer concentration and capex pacing, with 75.98% of FY25 sales in five anonymized accounts; (4) China/export policy and currency exposure. Sub-risks are grouped once to avoid double counting, not omitted. Capital-market scale does not resolve qualification risk. No unsupported customer-prepayment or monopoly bonus is awarded.

**Reference policy treatment.** The September 25 Senate proposal names Innolight, Eoptolink and related entities/components in restrictions on federal national-security procurement. Its sponsor describes a five-year transition and waivers [S21]. This is not an enacted general ban on commercial hyperscaler purchases. No unsupported immediate revenue loss is booked. The concrete trigger is broader legal scope or customer reallocation before implementation; either requires revisiting capture. Overseas assembly is not presumed to exempt a named company. This evidence is covered by the existing policy risk deduction, without adding a second deduction for the same risk class.

Micro-LED short-reach links are a longer-term technology watch, not evidence of an imminent collapse in 800G/1.6T laser demand. The user-supplied TrendForce USD848M 2030 micro-LED CPO forecast has a narrow market boundary and is not the entire optical market. No near-term blanket revenue haircut or new micro-LED revenue is included.

## 7. Probability ledger and pre-registered tests

| Program / test | Current treatment | Next evidence and counter |
|---|---|---|
| 800G/1.6T current programs | No extra probability haircut; actual revenue anchors continuation. Capture and allocation still estimated. | Q3 2026 filing, expected by end October: verify revenue, GM, cash and inventory. Revenue >10% below modeled quarter or GM <43% requires rebuilding. |
| H2 2027 advanced optics | 60% x raw allocated revenue, zero before Q3 2027. Timing from management; size is LOW. | Firm production order supports 75%; repeat volume shipments 90-95%. Two-quarter slip resets timing and probability to 35%. |
| 2028 advanced optics | 70% x raw pool; 75% in later years. No second TAM addition. | Require product-level disclosures and repeat customer deliveries. Lab demos alone do not justify a revenue increase. |
| Capacity and components | Projected module sales conditional on expanding capacity and securing input allocations. | Quarterly shipments outrunning feasible capacity, DSP allocations or yields cuts capture. Supplier prepayments are not proof of finished-goods delivery. |
| Cash quality | CFO-minus-capex tracked separately from OP ceiling. | Next two reports: inventory and supplier deposits must begin converting; continuing negative CFO with rising inventory is an execution counter. |
| Capex / architecture | Four-company basket monitored each reporting cycle. | Distinguish lease reclassification from physical spending cuts. Two independent real customer capex cuts trigger the framework review discipline. |

## 8. Sources, limits and artifact memory

Serenity is given the user's highest source priority; that priority does not change forecast versus reported status. The GS chart and the APAC screenshot are secondary evidence about one underlying report each, not multiple independent confirmations. Company filings control actual financial, share and manufacturing facts. No numeric company 2027 revenue guidance or customer-specific sales breakdown was found in the reviewed records. A current comparable Yahoo HIGH estimate was not verifiable and is not invented; the capex/actual-calibrated Plan-B model controls.

Remaining gaps: exact company ASPs and speed units, component allocations, detailed minority profit attribution, forward customer wallet shares and NPO contract amounts. These gaps make the forecast MEDIUM through 2027 and LOW thereafter. Subsequent evidence should replace the corresponding assumptions, not be layered on top of them.

- **S1 | 2026-08-22:** H1 2026 CAS report, pp.7-8, 16-22, 51-61 and finance notes.
- **S2 | 2026-04-17:** Q1 2026 CAS report, pp.2-3, 8-13.
- **S3 | 2026-03-31:** FY2025 annual report, pp.8-9,24-27 and consolidated statements.
- **S4 | 2026-08-23:** Company investor Q&A, full nine-page filing.
- **S5 | 2026-07-28:** Company investor Q&A, full six-page filing.
- **S6 | 2026-08-27:** Full exercise of H-share greenshoe; A + H capital.

- **S7 | 2026-09-25:** Repurchase completion; shares intended for employee awards.
- **S8 | 2026-09-28:** Company website equity-price snapshot.
- **S9 | 2026-09-08:** Serenity reporting GS optical TAM / ASP forecast; user-supplied source.
- **S10 | 2026-09-08:** Serenity GS ASP / CPO revision context and supplied chart.
- **S11 | 2026-09-25:** User-supplied P Equity / APAC conference screenshot; channel checks.
- **S12 | 2026-07-29:** Microsoft FY26 Q4 call: CY26 capex 175B, lease reclassification.
- **S13 | 2026-07-29:** Meta Q2 release: CY26 capex 130-145B.
- **S14 | 2026-07-22:** Reuters reporting Alphabet call: capex 195-205B; secondary transcript evidence.
- **S15 | 2026-07-30:** AP reporting Amazon call: 220B capex; secondary transcript evidence.
- **S16 | 2026:** Amazon 2025 annual report: cash capex 128.3B.
- **S17 | 2026-01:** Meta FY25 report: capex incl. lease principal 72.22B.
- **S18 | 2026:** Alphabet 2025 annual report, p.31: FY25 capex 91.4B.
- **S19 | 2025-04 to 2026-01:** Microsoft quarterly calls: CY25 capex 21.4 + 24.2 + 34.9 + 37.5 = 118.0B.
- **S20 | 2026-09-23:** Coherent PhotonLink: competing integration platform; not additive global TAM.
- **S21 | 2026-09-25:** Senate sponsor announcement: proposed optical-transceiver procurement restrictions.

Microsoft CY25 source calls also reviewed: FY25 Q3 | FY25 Q4 | FY26 Q1.

# Complete current operating model

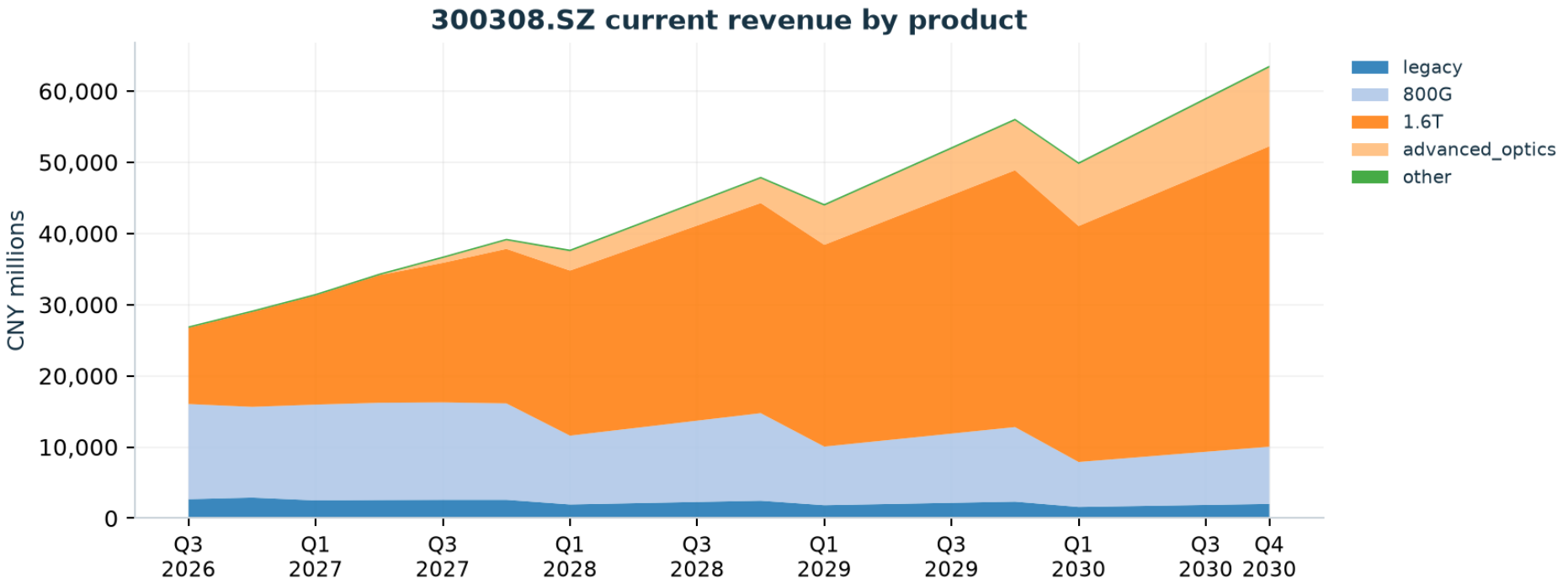



*Figure 4. Current quarterly revenue by product, CNY millions. All inherited product vectors remain; AAOI alone adds a separately labeled substitution vector.*

| Quarter | Revenue | Gross profit | COGS |
|---|---|---|---|
| Q3 2026 | 26,945.723 | 12,220.471 | 14,725.252 |
| Q4 2026 | 29,173.590 | 13,288.510 | 15,885.079 |
| Q1 2027 | 31,512.754 | 14,376.361 | 17,136.393 |
| Q2 2027 | 34,357.378 | 15,763.257 | 18,594.121 |
| Q3 2027 | 36,758.417 | 16,927.471 | 19,830.946 |
| Q4 2027 | 39,250.765 | 18,153.673 | 21,097.092 |
| Q1 2028 | 37,728.872 | 17,146.282 | 20,582.590 |
| Q2 2028 | 41,138.342 | 18,701.317 | 22,437.025 |

| Quarter | Revenue | Gross profit | COGS |
|---|---|---|---|
| Q3 2028 | 44,547.122 | 20,257.042 | 24,290.080 |
| Q4 2028 | 47,955.902 | 21,812.767 | 26,143.134 |
| Q1 2029 | 44,142.846 | 19,588.299 | 24,554.547 |
| Q2 2029 | 48,133.592 | 21,365.940 | 26,767.652 |
| Q3 2029 | 52,124.339 | 23,143.581 | 28,980.758 |
| Q4 2029 | 56,115.085 | 24,921.222 | 31,193.863 |
| Q1 2030 | 50,001.717 | 22,021.214 | 27,980.503 |
| Q2 2030 | 54,523.976 | 24,019.874 | 30,504.102 |
| Q3 2030 | 59,046.235 | 26,018.533 | 33,027.702 |
| Q4 2030 | 63,568.494 | 28,017.192 | 35,551.302 |

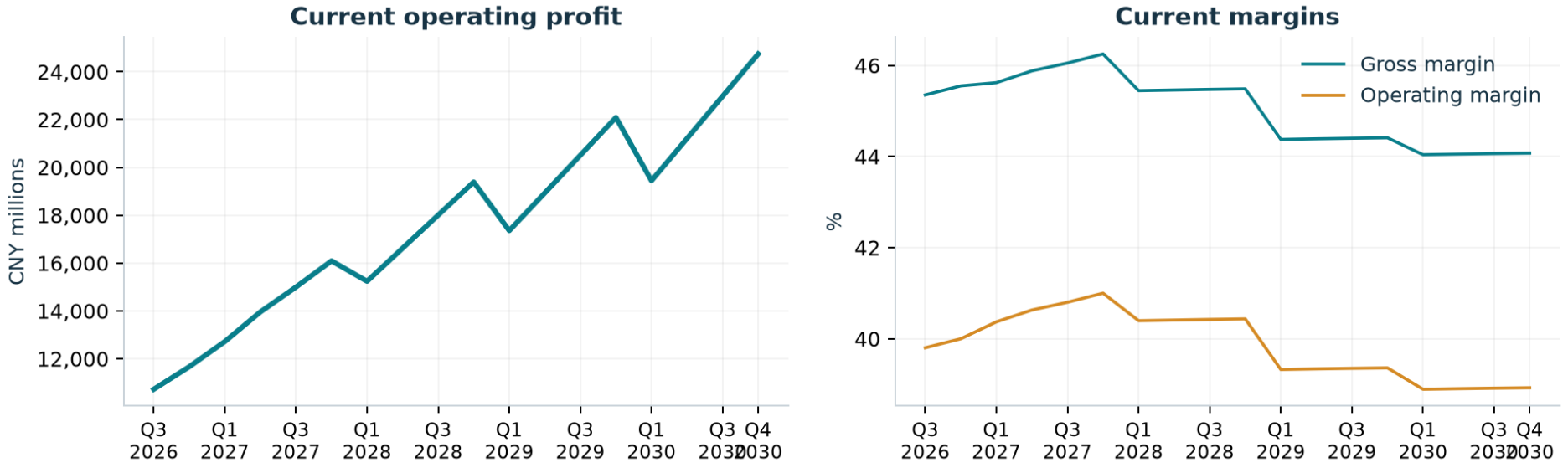



*Figure 5. Current operating profit and margins. Operating profit is consolidated CNY millions; the valuation uses parent-economic OP only where the inherited method already requires it.*

| Quarter | Revenue | Gross margin | Opex | Operating profit | OP margin | Valuation OP |
|---|---|---|---|---|---|---|
| Q3 2026 | 26,945.723 | 45.35% | 1,495.488 | 10,724.984 | 39.80% | 9,937.156 |
| Q4 2026 | 29,173.590 | 45.55% | 1,619.134 | 11,669.376 | 40.00% | 10,812.176 |
| Q1 2027 | 31,512.754 | 45.62% | 1,654.420 | 12,721.941 | 40.37% | 11,787.423 |
| Q2 2027 | 34,357.378 | 45.88% | 1,803.762 | 13,959.495 | 40.63% | 12,934.069 |
| Q3 2027 | 36,758.417 | 46.05% | 1,929.817 | 14,997.654 | 40.80% | 13,895.968 |
| Q4 2027 | 39,250.765 | 46.25% | 2,060.665 | 16,093.008 | 41.00% | 14,910.860 |
| Q1 2028 | 37,728.872 | 45.45% | 1,905.308 | 15,240.974 | 40.40% | 14,121.414 |
| Q2 2028 | 41,138.342 | 45.46% | 2,077.486 | 16,623.831 | 40.41% | 15,402.690 |
| Q3 2028 | 44,547.122 | 45.47% | 2,249.630 | 18,007.413 | 40.42% | 16,684.637 |
| Q4 2028 | 47,955.902 | 45.49% | 2,421.773 | 19,390.994 | 40.44% | 17,966.585 |
| Q1 2029 | 44,142.846 | 44.37% | 2,229.214 | 17,359.085 | 39.32% | 16,083.934 |
| Q2 2029 | 48,133.592 | 44.39% | 2,430.746 | 18,935.193 | 39.34% | 17,544.266 |
| Q3 2029 | 52,124.339 | 44.40% | 2,632.279 | 20,511.302 | 39.35% | 19,004.598 |
| Q4 2029 | 56,115.085 | 44.41% | 2,833.812 | 22,087.410 | 39.36% | 20,464.929 |
| Q1 2030 | 50,001.717 | 44.04% | 2,575.088 | 19,446.126 | 38.89% | 18,017.667 |
| Q2 2030 | 54,523.976 | 44.05% | 2,807.985 | 21,211.889 | 38.90% | 19,653.722 |
| Q3 2030 | 59,046.235 | 44.06% | 3,040.881 | 22,977.652 | 38.91% | 21,289.777 |
| Q4 2030 | 63,568.494 | 44.07% | 3,273.777 | 24,743.415 | 38.92% | 22,925.832 |

## Every revenue vector and product margin

Product GP = product revenue x the displayed modeled margin. Consolidated gross profit adds the unchanged source-model adjustment where applicable (including Sivers Glasgow depreciation). Product allocations and future margins remain LOW-confidence model assumptions unless specifically identified as historical or guided.

## legacy

| Quarter | Revenue | Gross margin | Gross profit |
|---|---|---|---|
| Q3 2026 | 2,673.440 | 30.00% | 802.032 |
| Q4 2026 | 2,896.227 | 30.00% | 868.868 |
| Q1 2027 | 2,503.269 | 28.00% | 700.915 |
| Q2 2027 | 2,560.162 | 28.00% | 716.845 |
| Q3 2027 | 2,588.608 | 28.00% | 724.810 |
| Q4 2027 | 2,588.608 | 28.00% | 724.810 |
| Q1 2028 | 1,932.671 | 26.00% | 502.494 |
| Q2 2028 | 2,108.368 | 26.00% | 548.176 |
| Q3 2028 | 2,284.066 | 26.00% | 593.857 |
| Q4 2028 | 2,459.763 | 26.00% | 639.538 |
| Q1 2029 | 1,828.859 | 25.00% | 457.215 |
| Q2 2029 | 1,995.119 | 25.00% | 498.780 |
| Q3 2029 | 2,161.379 | 25.00% | 540.345 |
| Q4 2029 | 2,327.639 | 25.00% | 581.910 |
| Q1 2030 | 1,578.915 | 24.00% | 378.940 |
| Q2 2030 | 1,722.453 | 24.00% | 413.389 |
| Q3 2030 | 1,865.990 | 24.00% | 447.838 |
| Q4 2030 | 2,009.528 | 24.00% | 482.287 |

## 800G

| Quarter | Revenue | Gross margin | Gross profit |
|---|---|---|---|
| Q3 2026 | 13,367.199 | 46.00% | 6,148.912 |
| Q4 2026 | 12,743.397 | 46.00% | 5,861.962 |
| Q1 2027 | 13,455.071 | 45.00% | 6,054.782 |
| Q2 2027 | 13,654.195 | 45.00% | 6,144.388 |
| Q3 2027 | 13,682.641 | 45.00% | 6,157.189 |
| Q4 2027 | 13,540.410 | 45.00% | 6,093.184 |
| Q1 2028 | 9,663.355 | 43.00% | 4,155.243 |
| Q2 2028 | 10,541.842 | 43.00% | 4,532.992 |
| Q3 2028 | 11,420.329 | 43.00% | 4,910.741 |
| Q4 2028 | 12,298.815 | 43.00% | 5,288.491 |
| Q1 2029 | 8,229.865 | 41.00% | 3,374.245 |
| Q2 2029 | 8,978.035 | 41.00% | 3,680.994 |
| Q3 2029 | 9,726.204 | 41.00% | 3,987.744 |
| Q4 2029 | 10,474.374 | 41.00% | 4,294.493 |
| Q1 2030 | 6,315.660 | 40.00% | 2,526.264 |
| Q2 2030 | 6,889.810 | 40.00% | 2,755.924 |
| Q3 2030 | 7,463.961 | 40.00% | 2,985.585 |
| Q4 2030 | 8,038.112 | 40.00% | 3,215.245 |

## 1.6T

| Quarter | Revenue | Gross margin | Gross profit |
|---|---|---|---|
| Q3 2026 | 10,693.759 | 49.00% | 5,239.942 |
| Q4 2026 | 13,322.642 | 49.00% | 6,528.095 |
| Q1 2027 | 15,332.523 | 49.50% | 7,589.599 |
| Q2 2027 | 17,921.131 | 49.50% | 8,870.960 |
| Q3 2027 | 19,599.459 | 49.50% | 9,701.732 |
| Q4 2027 | 21,704.481 | 49.50% | 10,743.718 |
| Q1 2028 | 23,192.052 | 48.00% | 11,132.185 |
| Q2 2028 | 25,300.420 | 48.00% | 12,144.202 |
| Q3 2028 | 27,408.788 | 48.00% | 13,156.218 |
| Q4 2028 | 29,517.157 | 48.00% | 14,168.235 |
| Q1 2029 | 28,347.314 | 46.00% | 13,039.764 |
| Q2 2029 | 30,924.342 | 46.00% | 14,225.197 |
| Q3 2029 | 33,501.371 | 46.00% | 15,410.630 |
| Q4 2029 | 36,078.399 | 46.00% | 16,596.064 |
| Q1 2030 | 33,157.213 | 45.00% | 14,920.746 |
| Q2 2030 | 36,171.505 | 45.00% | 16,277.177 |
| Q3 2030 | 39,185.797 | 45.00% | 17,633.608 |
| Q4 2030 | 42,200.089 | 45.00% | 18,990.040 |

## advanced_optics

| Quarter | Revenue | Gross margin | Gross profit |
|---|---|---|---|
| Q3 2026 | 0.000 | 47.00% | 0.000 |
| Q4 2026 | 0.000 | 47.00% | 0.000 |
| Q1 2027 | 0.000 | 47.00% | 0.000 |
| Q2 2027 | 0.000 | 47.00% | 0.000 |
| Q3 2027 | 665.818 | 46.96% | 312.675 |
| Q4 2027 | 1,195.375 | 46.92% | 560.896 |
| Q1 2028 | 2,707.809 | 48.89% | 1,323.742 |
| Q2 2028 | 2,954.726 | 48.85% | 1,443.330 |
| Q3 2028 | 3,200.954 | 48.85% | 1,563.607 |
| Q4 2028 | 3,447.181 | 48.85% | 1,683.885 |
| Q1 2029 | 5,492.173 | 48.85% | 2,682.826 |
| Q2 2029 | 5,991.462 | 48.85% | 2,926.720 |
| Q3 2029 | 6,490.750 | 48.85% | 3,170.613 |
| Q4 2029 | 6,990.039 | 48.85% | 3,414.506 |
| Q1 2030 | 8,693.063 | 47.85% | 4,159.304 |
| Q2 2030 | 9,483.342 | 47.85% | 4,537.422 |
| Q3 2030 | 10,273.620 | 47.85% | 4,915.541 |
| Q4 2030 | 11,063.899 | 47.85% | 5,293.660 |

### other

| Quarter | Revenue | Gross margin | Gross profit |
|---|---|---|---|
| Q3 2026 | 211.325 | 14.00% | 29.585 |
| Q4 2026 | 211.325 | 14.00% | 29.585 |
| Q1 2027 | 221.891 | 14.00% | 31.065 |
| Q2 2027 | 221.891 | 14.00% | 31.065 |
| Q3 2027 | 221.891 | 14.00% | 31.065 |
| Q4 2027 | 221.891 | 14.00% | 31.065 |
| Q1 2028 | 232.986 | 14.00% | 32.618 |
| Q2 2028 | 232.986 | 14.00% | 32.618 |
| Q3 2028 | 232.986 | 14.00% | 32.618 |
| Q4 2028 | 232.986 | 14.00% | 32.618 |
| Q1 2029 | 244.635 | 14.00% | 34.249 |
| Q2 2029 | 244.635 | 14.00% | 34.249 |
| Q3 2029 | 244.635 | 14.00% | 34.249 |
| Q4 2029 | 244.635 | 14.00% | 34.249 |
| Q1 2030 | 256.867 | 14.00% | 35.961 |
| Q2 2030 | 256.867 | 14.00% | 35.961 |
| Q3 2030 | 256.867 | 14.00% | 35.961 |
| Q4 2030 | 256.867 | 14.00% | 35.961 |

## Full ceiling derivation

Valuation OP equals consolidated OP except where the saved model allocates economics to parent shareholders. The current model adds ownership x incremental OP to the saved parent-economic OP. Existing parent-only expenses and senior claims remain deducted as before. For every ceiling, sum that quarter and its next three quarters, multiply by the sanctioned multiple and currency/unit factor, and divide by the saved quarter-specific share denominator. No stock-price-dependent upside or ranking appears in this artifact.

| Ceiling quarter / window | F4Q valuation OP | Multiple | Shares M | FX / unit factor | Ceiling |
|---|---|---|---|---|---|
| Q3 2026 to Q2 2027 | 45,470.823167 | 20x | 1,190.000000 | 1.000000 | 764.215515 |
| Q4 2026 to Q3 2027 | 49,429.635050 | 20x | 1,190.000000 | 1.000000 | 830.750169 |
| Q1 2027 to Q4 2027 | 53,528.318801 | 20x | 1,190.000000 | 1.000000 | 899.635610 |
| Q2 2027 to Q1 2028 | 55,862.310302 | 20x | 1,190.000000 | 1.000000 | 938.862358 |
| Q3 2027 to Q2 2028 | 58,330.931344 | 20x | 1,190.000000 | 1.000000 | 980.351787 |
| Q4 2027 to Q3 2028 | 61,119.601176 | 20x | 1,190.000000 | 1.000000 | 1,027.220188 |
| Q1 2028 to Q4 2028 | 64,175.326460 | 20x | 1,190.000000 | 1.000000 | 1,078.576915 |
| Q2 2028 to Q1 2029 | 66,137.846826 | 20x | 1,190.000000 | 1.000000 | 1,111.560451 |
| Q3 2028 to Q2 2029 | 68,279.422930 | 20x | 1,190.000000 | 1.000000 | 1,147.553327 |
| Q4 2028 to Q3 2029 | 70,599.383207 | 20x | 1,190.000000 | 1.000000 | 1,186.544256 |
| Q1 2029 to Q4 2029 | 73,097.727658 | 20x | 1,190.000000 | 1.000000 | 1,228.533238 |
| Q2 2029 to Q1 2030 | 75,031.460013 | 20x | 1,190.000000 | 1.000000 | 1,261.032941 |
| Q3 2029 to Q2 2030 | 77,140.915644 | 20x | 1,190.000000 | 1.000000 | 1,296.485977 |
| Q4 2029 to Q3 2030 | 79,426.094549 | 20x | 1,190.000000 | 1.000000 | 1,334.892345 |
| Q1 2030 to Q4 2030 | 81,886.996730 | 20x | 1,190.000000 | 1.000000 | 1,376.252046 |

**Q4 2027 worked example:** (14,910.859631 + 14,121.414097 + 15,402.689993 + 16,684.637454) x 20 x 1 / 1190.000000 = **1,027.220188 CNY/share**.

## Annual cross-check

Calendar-year sums below include only the quarters in the saved model. Partial years are labeled. These are not additional revenue, a horizon roll, or new management guidance.

| Calendar year | Coverage | Revenue | Gross profit | Operating profit | OP margin |
|---|---|---|---|---|---|
| 2026 | 2 quarters | 56,119.313 | 25,508.982 | 22,394.360 | 39.90% |
| 2027 | 4 quarters | 141,879.315 | 65,220.763 | 57,772.099 | 40.72% |
| 2028 | 4 quarters | 171,370.238 | 77,917.409 | 69,263.212 | 40.42% |
| 2029 | 4 quarters | 200,515.861 | 89,019.041 | 78,892.990 | 39.35% |

| Calendar year | Coverage | Revenue | Gross profit | Operating profit | OP margin |
|---|---|---|---|---|---|
| 2030 | 4 quarters | 227,140.422 | 100,076.813 | 88,379.081 | 38.91% |

## Probability and dilution sensitivity

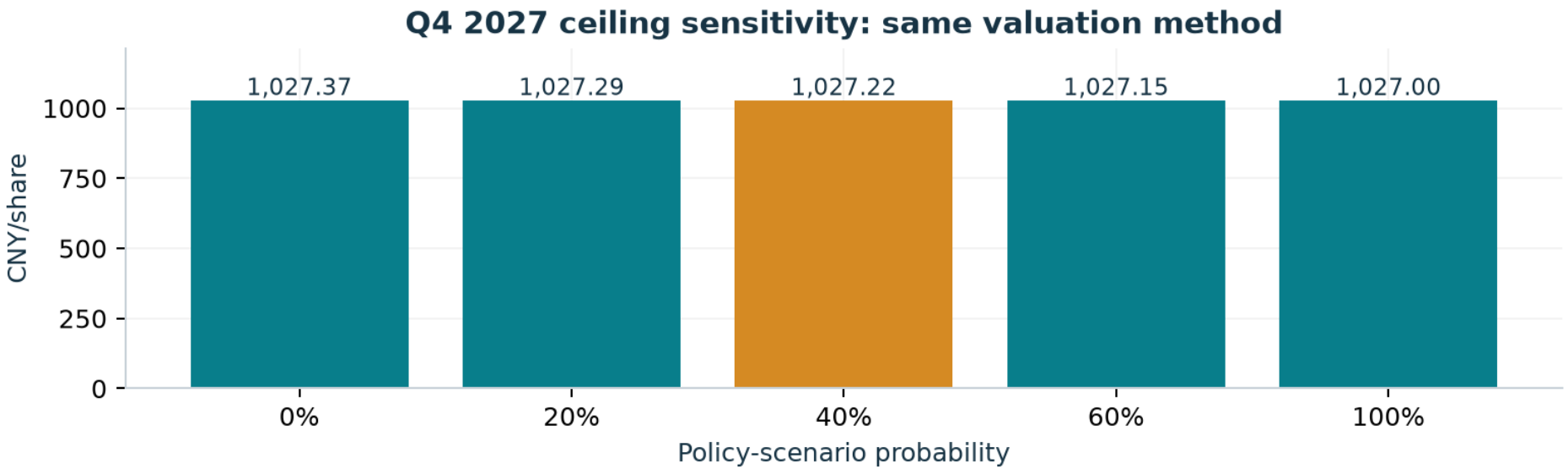



*Figure 6. Q4 2027 ceiling under alternative policy probabilities. Gold is the selected 40% weight. All company valuation methods, operating reference inputs and shares remain unchanged.*

| Policy probability | Q4 2027 ceiling | Change from zero-policy sensitivity |
|---|---|---|
| 0% | 1,027.3670 | 0.000% |
| 20% | 1,027.2936 | -0.007% |
| 40% | 1,027.2202 | -0.014% |
| 60% | 1,027.1468 | -0.021% |
| 100% | 1,026.9999 | -0.036% |

| Additional diluted shares | Q4 2027 Ceiling | Interpretation |
|---|---|---|
| 0% | 1,027.2202 | Same operating case; denominator sensitivity only |

| Additional diluted shares | Q4 2027 Ceiling | Interpretation |
|---|---|---|
| 5% | 978.3049 | Same operating case; denominator sensitivity only |
| 10% | 933.8365 | Same operating case; denominator sensitivity only |

The 40% weight is a judgment about the incremental policy economics and timing, not a statistical certainty. Because the nine companies share the same policy event, these increments should not be treated as nine independent sources of portfolio diversification. The next 12-16 months have the highest forecasting relevance; later continuation assumptions are progressively less reliable.

## Validation and next evidence

This current artifact and the master model use the same selected 40% policy case. Existing company valuation methods, quarter conventions, ownership and dilution schedules are retained. Every existing portfolio Excel Base is protected; the analytical score above is assessed separately.

The incremental ledger exposes revenue, production costs, gross profit and operating expense. A zero-policy sensitivity removes only that ledger's contribution. It is a diagnostic control, not a competing active forecast. Reported facts and dated guidance retain their original dates; new pricing, exposure, costs and timing are labeled analyst assumptions.

The most useful next evidence is the proposed content definition, product-specific fabrication and assembly routes, contract repricing rights, delivered 3.2T qualification, accepted component premiums and unit-cost/yield data. Those observations can raise or lower the inputs and their probability; official confirmation is not a prerequisite for modeling this credible breadcrumb.

Current model: **300308_SZ_Q_Model** in **v3_2_master_portfolio.xlsx**. The full quarterly model includes all original product vectors and the approved incremental operating bridge. Approval snapshots and change history are retained externally rather than as competing current forecasts.